
Chapter 5: Trigonometry

A Level Pure Maths Notes · Rexponential

Why this chapter matters

You've already met right-angled trigonometry, the sine rule, cosine rule, and probably radians in some form. A level trig takes those ideas much further: you'll work fluently in **radians** rather than degrees for calculus later on, learn the **reciprocal** functions (sec, cosec, cot) and **inverse** functions (arcsin, arccos, arctan), and — the genuinely new material — the **addition formulae**, **double-angle formulae**, and the **R sin/cos** technique for combining trig terms. These culminate in Chapter 7 and 8, where trig functions get differentiated and integrated, so real fluency here (not just "I can look it up") pays off later.

A note on units: A level Maths uses both degrees and radians depending on context, but as soon as calculus is involved (Chapters 7–8), radians are compulsory — the derivative formulas for sin and cos only work in radians. Get comfortable converting between the two now.

5.1 The sine rule, cosine rule, and area of a triangle

Concept

The sine rule and cosine rule extend right-angled trigonometry (SOHCAHTOA) to *any* triangle. The sine rule relates each side to the sine of its opposite angle; the cosine rule is a generalisation of Pythagoras that accounts for the angle between two known sides not being 90° . Which one you need depends on what information you're given — this is a recognition skill as much as a computational one.

Procedure

- **Sine rule:** $a/\sin A = b/\sin B = c/\sin C$. Use when you know a side and its opposite angle, plus one more piece of information (either "angle-angle-side" or "side-side-angle," the latter needing care over the ambiguous case).
- **Cosine rule:** $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ (and cyclically for b^2 , c^2). Use when you know all three sides (to find an angle — rearrange for $\cos A$) or two sides and the *included* angle (to find the third side).
- **Area of a triangle:** $\text{Area} = \frac{1}{2}ab \cdot \sin C$, using two sides and the included angle — far more useful than $\frac{1}{2} \times \text{base} \times \text{height}$ when you don't have a perpendicular height to hand.

Worked example

EXAMPLE 5.1

(a) In triangle ABC, $a = 8$, $B = 40^\circ$, $C = 65^\circ$. Find b . (b) In triangle ABC, $a = 7$, $b = 9$, $C = 55^\circ$. Find c and the area of the triangle. (c) In triangle ABC, $a = 5$, $b = 7$, $c = 10$. Find the largest angle.

(a) First find $A = 180^\circ - 40^\circ - 65^\circ = 75^\circ$ (angles in a triangle sum to 180°) (M1). By the sine rule: $b/\sin 40^\circ = 8/\sin 75^\circ \Rightarrow b = 8 \sin 40^\circ / \sin 75^\circ = \mathbf{5.32}$ (3 s.f.) (M1, A1).

(b) Since we know two sides and the included angle, use the cosine rule directly: $c^2 = 7^2 + 9^2 - 2(7)(9)\cos 55^\circ = 49 + 81 - 126\cos 55^\circ = 57.7\dots \Rightarrow c = \mathbf{7.60}$ (3 s.f.) (M1, A1). Area = $\frac{1}{2}(7)(9)\sin 55^\circ = \mathbf{25.8}$ (3 s.f.) (M1, A1).

(c) The largest angle is opposite the longest side, so we want angle C (opposite $c = 10$). Rearranging the cosine rule (M1): $\cos C = (a^2 + b^2 - c^2)/(2ab) = (25 + 49 - 100)/(2 \times 5 \times 7) = -26/70 = -0.371\dots$ $C = \cos^{-1}(-0.371\dots) = \mathbf{111.8^\circ}$ (3 s.f.) (A1). (Note the negative cosine value — that's your signal the angle is obtuse, which makes sense since it's the largest angle in the triangle.)

Answer (a) $b = 5.32$ (b) $c = 7.60$, area = 25.8 (c) largest angle = 111.8° .

EXAM TIP

Using the sine rule when the cosine rule is needed (or vice versa) — before starting, always label what you know (sides/angles) and check whether you have an angle *opposite* a known side (sine rule) or need to relate three sides plus one angle where the angle isn't opposite the unknown (cosine rule).

5.2 Radians: arc length, sector and segment area

Concept

A radian is defined so that an arc of length equal to the radius subtends an angle of exactly 1 radian at the centre. This definition is what makes radians the "natural" unit for angles in calculus — formulas for arc length and sector area become clean multiplications rather than needing a 360° -based fraction. Conversion: $180^\circ = \pi$ radians, so to convert degrees $\rightarrow$ radians multiply by $\pi/180$, and radians $\rightarrow$ degrees multiply by $180/\pi$.

Procedure

For a sector of a circle with radius r and angle θ **in radians**:

- Arc length: $s = r\theta$
- Sector area: $A = \frac{1}{2}r^2\theta$
- Segment area (the region between a chord and the arc): $A = \frac{1}{2}r^2(\theta - \sin\theta)$ — this is sector area minus the triangle area $\frac{1}{2}r^2\sin\theta$ formed by the two radii and the chord.

These formulas **ONLY** work with θ in radians — if a question gives an angle in degrees, convert first.

Worked example

EXAMPLE 5.2

A sector of a circle has radius 6 cm and angle 1.2 radians. Find (a) the arc length, (b) the sector area, (c) the area of the segment cut off by the chord.

(a) $s = r\theta = 6 \times 1.2 = 7.2 \text{ cm}$ (M1, A1).

(b) $A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)(1.2) = 21.6 \text{ cm}^2$ (M1, A1).

(c) Segment area $= \frac{1}{2}r^2(\theta - \sin\theta) = \frac{1}{2}(36)(1.2 - \sin(1.2)) = 18 \times (1.2 - 0.932...) = 18 \times 0.268... = 4.82 \text{ cm}^2$ (3 s.f.) (M1, A1). (Note: $\sin(1.2)$ here means $\sin$ of 1.2 radians — make sure your calculator is in radian mode.)

Answer (a) 7.2 cm (b) 21.6 cm² (c) 4.82 cm².

EXAM TIP

Using a degree-mode calculator, or a degrees-based formula, with an angle already given in radians (or vice versa) — always check your calculator's angle mode before starting, and sanity check: an angle in radians rarely exceeds about 6.3 (2π), so a "radian" answer of, say, 55 is almost certainly a degrees/radians mix-up.

5.3 Graphs, exact values, and solving trig equations

Concept

The graphs of $y = \sin\theta$, $y = \cos\theta$, and $y = \tan\theta$ are periodic — they repeat forever. $\sin$ and $\cos$ both have period 360° (2π radians) and oscillate between -1 and 1 ; $\tan$ has period 180° (π radians) and is unbounded, with vertical asymptotes wherever $\cos\theta = 0$ (i.e. at 90° , 270° , ...). Because the functions repeat, a trig **equation** (as opposed to just evaluating $\sin$ of something) generally has infinitely many solutions — exam questions restrict you to a given range, and finding *all* solutions in that range (not just one) is the actual skill being tested.

EXACT VALUES WORTH MEMORISING (FROM THE 30-60-90 AND 45-45-90 TRIANGLES)

θ	0°	30°	45°	60°	90°
$\sin\theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan\theta$	0	$\sqrt{3}/3$	1	$\sqrt{3}$	undefined

Procedure for solving trig equations

1. Isolate the trig function (e.g. rearrange to $\sin x = k$).
2. Find the *principal value* using the inverse function (see 5.6).
3. Use the graph's symmetry (or the CAST diagram, if you use that convention) to find all other solutions in the given range — e.g. for $\sin x = k$ with a positive k , solutions are at $x = \text{principal value}$ and $x = 180^\circ - \text{principal value}$ (plus/minus multiples of 360° to land inside the range).

4. List every solution the range asks for, in order.

Worked example

EXAMPLE 5.3

Solve $2 \sin x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

Step 1 Rearranging: $\sin x = 1/2$ (M1). Principal value: $x = \sin^{-1}(1/2) = 30^\circ$ (A1).

Step 2 Since sine is positive in the first *and* second quadrants, the second solution is $x = 180^\circ - 30^\circ = 150^\circ$ (M1). Both lie within $[0^\circ, 360^\circ]$, so the solutions are **$x = 30^\circ$ or $x = 150^\circ$** (A1). (*Always check: are there more solutions further round? Adding 360° to either would leave the range, so these two are all of them.*)

Answer $x = 30^\circ$ or $x = 150^\circ$.

EXAM TIP

Finding only the principal-value solution and stopping there — always ask "where else, within the graph's period, does the function take this value?" before finalising an answer.

5.4 The Identity $\sin^2\theta + \cos^2\theta = 1$

Concept

$\sin^2\theta + \cos^2\theta = 1$ is just Pythagoras' theorem applied to the unit circle: if a point on the unit circle is at angle θ , its coordinates are $(\cos\theta, \sin\theta)$, and since the point is distance 1 from the origin, $\cos^2\theta + \sin^2\theta = 1^2 = 1$ follows directly. $\tan\theta = \sin\theta/\cos\theta$ is essentially the definition of $\tan$ on the unit circle (rise over run, expressed via $\sin$ and $\cos$). These two identities are the foundation every other trig identity in this course is built from.

Procedure

Use $\sin^2\theta + \cos^2\theta = 1$ to:

- find $\cos\theta$ given $\sin\theta$ (or vice versa), being careful about the sign, which depends on which quadrant θ is in;
- simplify or prove trigonometric identities by substituting $\sin^2\theta = 1 - \cos^2\theta$ (or the reverse) to eliminate one function in favour of the other;
- solve equations that mix $\sin^2$ and $\cos^2$ terms by rewriting everything in terms of a single function.

Worked example

EXAMPLE 5.4

Given that $\sin\theta = 3/5$ and θ is acute, find the exact values of $\cos\theta$ and $\tan\theta$.

Step 1 From the identity: $\cos^2\theta = 1 - \sin^2\theta = 1 - 9/25 = 16/25$ (M1).

Step 2 Since θ is acute, $\cos\theta$ is positive, so $\cos\theta = \sqrt{16/25} = 4/5$ (A1). *(This is the classic 3-4-5 triangle — recognising it can save time, but the identity method always works even when the numbers aren't as clean.)*

Step 3 $\tan\theta = \sin\theta/\cos\theta = (3/5)/(4/5) = 3/4$ (M1, A1).

Answer $\cos\theta = 4/5$, $\tan\theta = 3/4$.

EXAM TIP

Forgetting to consider the sign of $\cos\theta$ (or $\sin\theta$) when θ is not stated to be acute — if θ could be in the second quadrant, for instance, $\cos\theta$ would be negative even though $\sin^2\theta$ is unchanged; always use the given information about θ 's range to fix the sign.

5.5 The reciprocal functions: sec, cosec, cot

Concept

sec, cosec (sometimes written csc), and cot are simply the reciprocals of cos, sin, and tan respectively — they don't carry any new geometric meaning beyond "1 divided by." They exist mainly because certain identities and integrals (especially in Chapters 7–8) are tidier written this way. The names are a useful memory aid: **sec**ant pairs with cos, **cosec**ant pairs with sin (note the *swap* — sec goes with cos, cosec goes with sin, which trips people up at first), and **cot**angent pairs with tan.

Procedure

KEY FORMULAS: RECIPROCAL FUNCTIONS

- $\sec\theta = 1/\cos\theta$
- $\operatorname{cosec}\theta = 1/\sin\theta$
- $\cot\theta = 1/\tan\theta = \cos\theta/\sin\theta$
- $1 + \tan^2\theta = \sec^2\theta$ (dividing $\sin^2\theta + \cos^2\theta = 1$ through by $\cos^2\theta$)
- $1 + \cot^2\theta = \operatorname{cosec}^2\theta$ (dividing $\sin^2\theta + \cos^2\theta = 1$ through by $\sin^2\theta$)

Because these are reciprocals, $\sec\theta$ and $\operatorname{cosec}\theta$ are undefined wherever $\cos\theta = 0$ or $\sin\theta = 0$ respectively, and $|\sec\theta| \geq 1$, $|\operatorname{cosec}\theta| \geq 1$ always (since $|\cos\theta|, |\sin\theta| \leq 1$).

Worked example

EXAMPLE 5.5

Find the exact values of $\sec 60^\circ$, $\operatorname{cosec} 60^\circ$, and $\cot 60^\circ$.

Step 1 $\cos 60^\circ = 1/2$, so $\sec 60^\circ = 1/(1/2) = 2$ (B1).

Step 2 $\sin 60^\circ = \sqrt{3}/2$, so $\operatorname{cosec} 60^\circ = 1/(\sqrt{3}/2) = 2/\sqrt{3} = 2\sqrt{3}/3$ (B1, rationalised).

Step 3 $\tan 60^\circ = \sqrt{3}$, so $\cot 60^\circ = 1/\sqrt{3} = \sqrt{3}/3$ (B1, rationalised).

Answer $\sec 60^\circ = 2$, $\operatorname{cosec} 60^\circ = 2\sqrt{3}/3$, $\cot 60^\circ = \sqrt{3}/3$.

EXAM TIP

Confusing $\sec$ with $1/\sin$ (i.e. muddling $\sec$ and cosec) — the mismatched-letter pairing ($\sec \leftrightarrow \cos$, $\operatorname{cosec} \leftrightarrow \sin$) is genuinely counterintuitive, so it's worth deliberately memorising rather than assuming it follows the obvious pattern.

5.6 Inverse trig functions

Concept

$\sin$, $\cos$, and $\tan$ are not one-to-one over all real numbers (they repeat), so they don't have a true inverse function on their full domain. To define $\arcsin$, $\arccos$, and $\arctan$ properly, we **restrict the domain** of $\sin$, $\cos$, $\tan$ to an interval where each is one-to-one, and the inverse function is defined only on that restricted piece. This is why $\arcsin$ and $\arccos$ only ever output values in a limited range — it's not a limitation of your calculator, it's a mathematical necessity for the inverse to be well-defined.

Procedure

DOMAIN AND RANGE OF THE INVERSE TRIG FUNCTIONS

- $\arcsin (\sin^{-1})$: domain $[-1, 1]$, range $[-90^\circ, 90^\circ]$ (or $[-\pi/2, \pi/2]$ in radians).
- $\arccos (\cos^{-1})$: domain $[-1, 1]$, range $[0^\circ, 180^\circ]$ (or $[0, \pi]$).
- $\arctan (\tan^{-1})$: domain all reals, range $(-90^\circ, 90^\circ)$ (or $(-\pi/2, \pi/2)$), not including the endpoints since $\tan$ is undefined there.

The value your calculator gives when you press $\sin^{-1}$, $\cos^{-1}$, or $\tan^{-1}$ is always this **principal value**, inside the ranges above — for an equation with solutions outside that range, you must use the symmetry of the graph (as in 5.3) to find them, the calculator won't do it for you.

Worked example

EXAMPLE 5.6

Find the exact values of $\arcsin(1/2)$, $\arccos(-1/2)$, and $\arctan(1)$.

Step 1 $\arcsin(1/2)$: we need an angle in $[-90^\circ, 90^\circ]$ whose sine is $1/2$ — that's **30°** (A1).

Step 2 $\arccos(-1/2)$: we need an angle in $[0^\circ, 180^\circ]$ whose cosine is $-1/2$. Since cosine is negative, the angle is obtuse: **120°** (A1). ($\cos 60^\circ = 1/2$, and by the graph's symmetry $\cos(180^\circ - 60^\circ) = -\cos 60^\circ = -1/2$.)

Step 3 $\arctan(1)$: we need an angle in $(-90^\circ, 90^\circ)$ whose tangent is 1 — that's **45°** (A1).

Answer $\arcsin(1/2) = 30^\circ$, $\arccos(-1/2) = 120^\circ$, $\arctan(1) = 45^\circ$.

EXAM TIP

Giving $\arccos$ of a negative number as a negative angle (by analogy with $\arcsin$) — remember $\arccos$ always returns a value in $[0^\circ, 180^\circ]$, never negative, precisely because $\cos$'s domain was restricted differently from $\sin$'s.

5.7 Addition formulae

Concept

The addition formulae express $\sin$, $\cos$, and $\tan$ of a *sum or difference* of two angles in terms of the sines, cosines, and tangents of the individual angles. They are not the same as " $\sin(A+B) = \sin A + \sin B$ " — that's a very common and completely wrong shortcut. The formulae come from combining rotations on the unit circle and, while you're not usually asked to prove them at this level, you'll use them constantly: to find exact values of unfamiliar angles (like $75^\circ = 45^\circ + 30^\circ$), to prove other identities, and as the building blocks for the double-angle formulae in 5.8.

Procedure

Learn these (they will be given in a formula sheet in some exam boards, but not all — check your specification):

KEY FORMULAS: ADDITION FORMULAE

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ (*note the flipped sign — this is the one everyone gets wrong first*)
- $\tan(A \pm B) = (\tan A \pm \tan B) / (1 \mp \tan A \tan B)$

To use them: split the angle you're given into two angles whose exact trig values you know (commonly using 30° , 45° , 60° , 90°), substitute into the formula, then simplify using exact values.

Worked example

EXAMPLE 5.7

Find the exact value of $\sin 75^\circ$.

Step 1 Write $75^\circ = 45^\circ + 30^\circ$ (M1 — chosen because both 45° and 30° have known exact values).

Step 2 $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$ (M1, using the addition formula).

Step 3 $= (\sqrt{2}/2)(\sqrt{3}/2) + (\sqrt{2}/2)(1/2) = \sqrt{6}/4 + \sqrt{2}/4 = (\sqrt{6} + \sqrt{2})/4$ (A1).

Answer $\sin 75^\circ = (\sqrt{6} + \sqrt{2})/4$.

EXAM TIP

Writing $\cos(A+B) = \cos A \cos B - \sin A \sin B$ (using the same sign as for $\sin$) — the sign flips specifically for cosine; it can help to remember it as "cos: same signs become opposite, opposite signs become same."

5.8 Double-angle formulae

Concept

The double-angle formulae are simply the addition formulae with $B = A$ — i.e. what happens when you find $\sin$, $\cos$, or $\tan$ of *twice* an angle in terms of the trig functions of the original angle. Because $\cos$ has three equivalent double-angle forms (using the $\sin^2 + \cos^2 = 1$ identity to substitute), you get to choose whichever form is most convenient for the problem at hand — that flexibility is exactly why $\cos 2A$ is quoted three ways.

Procedure

From $\sin(A+A)$ and $\cos(A+A)$:

KEY FORMULAS: DOUBLE-ANGLE FORMULAE

- $\sin 2A = 2 \sin A \cos A$
- $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$ (all three are equivalent; pick whichever matches what you're given — e.g. if you know $\sin A$, use the $1 - 2\sin^2 A$ form to avoid finding $\cos A$ first)
- $\tan 2A = 2\tan A / (1 - \tan^2 A)$

These are used constantly for simplifying expressions, solving equations involving both θ and 2θ , and (in reverse) for integrating $\sin^2 x$ and $\cos^2 x$ in Chapter 8, where the $1 \pm 2\sin^2/\cos^2$ forms let you rewrite a squared trig function as a single, integrable term.

Worked example

EXAMPLE 5.8

Given that $\sin\theta = 3/5$ and θ is acute, find the exact values of $\sin 2\theta$ and $\cos 2\theta$.

Step 1 Since θ is acute, from 5.4's method, $\cos\theta = 4/5$ (M1, reusing the identity $\sin^2 + \cos^2 = 1$).

Step 2 $\sin 2\theta = 2 \sin\theta \cos\theta = 2(3/5)(4/5) = 24/25 = \mathbf{0.96}$ (M1, A1).

Step 3 $\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2(9/25) = 1 - 18/25 = 7/25 = \mathbf{0.28}$ (M1, A1). (Using the $1 - 2\sin^2\theta$ form avoided needing $\cos\theta$ again — a small efficiency worth noticing.)

Answer $\sin 2\theta = 0.96$, $\cos 2\theta = 0.28$.

EXAM TIP

Writing $\sin 2A = 2 \sin A$ (dropping the $\cos A$ factor entirely) — there is no valid simplification of $\sin(2A)$ that doesn't involve both $\sin A$ and $\cos A$; always use the full formula.

5.9 The $R \sin(\theta \pm \alpha)$ / $R \cos(\theta \pm \alpha)$ form

Concept

An expression like $a \sin\theta + b \cos\theta$ (a sum of a sine and a cosine term with different amplitudes) can always be rewritten as a single sinusoidal wave $R \sin(\theta \pm \alpha)$, for some amplitude R and phase shift α . This matters because a single sine or cosine term is far easier to analyse — its maximum/minimum values and the equations it satisfies are immediate once it's in this form, whereas a sum of two separate trig terms isn't.

Procedure

To write $a \sin\theta + b \cos\theta$ as $R \sin(\theta + \alpha)$ (with $R > 0$ and $0^\circ < \alpha < 90^\circ$ assuming $a, b > 0$):

1. Expand $R \sin(\theta + \alpha) = R \sin\theta \cos\alpha + R \cos\theta \sin\alpha$ (using the addition formula) and compare coefficients with $a \sin\theta + b \cos\theta$: $R \cos\alpha = a$ and $R \sin\alpha = b$.
2. Square and add: $R^2 \cos^2\alpha + R^2 \sin^2\alpha = a^2 + b^2 \implies R^2 = a^2 + b^2 \implies \mathbf{R = \sqrt{a^2 + b^2}}$.
3. Divide: $(R \sin\alpha)/(R \cos\alpha) = \tan\alpha = b/a \implies \mathbf{\alpha = \tan^{-1}(b/a)}$.

Once in this form, $a \sin\theta + b \cos\theta$ has maximum value R (when $\theta + \alpha = 90^\circ$) and minimum $-R$, and equations like $a \sin\theta + b \cos\theta = k$ become straightforward to solve by first solving for $(\theta + \alpha)$ as a single unknown.

Worked example

EXAMPLE 5.9

(a) Express $3 \sin \theta + 4 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. (b) Hence solve $3 \sin \theta + 4 \cos \theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$.

(a) $R = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ (B1). $\alpha = \tan^{-1}(4/3) = 53.1^\circ$ (1 d.p.) (M1, A1). So $3 \sin \theta + 4 \cos \theta = 5 \sin(\theta + 53.1^\circ)$ (A1).

(b) $5 \sin(\theta + 53.1^\circ) = 2 \implies \sin(\theta + 53.1^\circ) = 0.4$ (M1). Let $\phi = \theta + 53.1^\circ$. Principal value: $\phi = \sin^{-1}(0.4) = 23.6^\circ$, second solution $\phi = 180^\circ - 23.6^\circ = 156.4^\circ$ (M1). Since θ ranges over $[0^\circ, 360^\circ]$, $\phi = \theta + 53.1^\circ$ ranges over $[53.1^\circ, 413.1^\circ]$, so we need every solution of $\sin \phi = 0.4$ in that wider range: $\phi = 156.4^\circ$ and $\phi = 23.6^\circ + 360^\circ = 383.6^\circ$ both lie in $[53.1^\circ, 413.1^\circ]$ (the first $\phi = 23.6^\circ$ itself does not, since it's below 53.1°) (M1). $\theta = \phi - 53.1^\circ$, so $\theta = 156.4^\circ - 53.1^\circ = 103.3^\circ$, or $\theta = 383.6^\circ - 53.1^\circ = 330.4^\circ$ (A1, both to 1 d.p.).

Answer (a) $3 \sin \theta + 4 \cos \theta = 5 \sin(\theta + 53.1^\circ)$ (b) $\theta = 103.3^\circ$ or $\theta = 330.4^\circ$.

EXAM TIP

Solving for $\phi = \theta + \alpha$ but forgetting to widen the search range accordingly (and then to shift the final answers back by $-\alpha$) — shifting the range is the step most often skipped, and it silently drops valid solutions.

5.10 Small-angle approximations

Concept

For small angles θ (measured **in radians**), $\sin \theta$, $\cos \theta$, and $\tan \theta$ can be approximated by simple polynomials in θ : $\sin \theta \approx \theta$, $\cos \theta \approx 1 - \theta^2/2$, $\tan \theta \approx \theta$. These come from the early terms of the Maclaurin series for each function (which you may meet formally at university), but at A level you're expected to use them as given approximations, verify them numerically, and apply them to simplify limits and physics-style problems where an angle is known to be small.

Procedure

For θ small and in radians:

KEY FORMULAS: SMALL-ANGLE APPROXIMATIONS

- $\sin \theta \approx \theta$
- $\cos \theta \approx 1 - \theta^2/2$
- $\tan \theta \approx \theta$ (since $\tan \theta = \sin \theta / \cos \theta \approx \theta / 1 = \theta$, to this level of approximation)

These are used to simplify expressions or limits as $\theta \rightarrow 0$, and to solve equations approximately when an exact solution is hard to find but θ is known to be small.

Worked example

EXAMPLE 5.10

Use the small-angle approximations to find an approximate value for $\sin(0.1)$ and $\cos(0.1)$, and compare with the calculator values.

Step 1 $\sin(0.1) \approx 0.1$ (using $\sin \theta \approx \theta$) (M1, A1). Calculator value: $\sin(0.1) = 0.09983\dots$ — agreement to 3 significant figures.

Step 2 $\cos(0.1) \approx 1 - (0.1)^2/2 = 1 - 0.005 = 0.995$ (M1, A1). Calculator value: $\cos(0.1) = 0.99500\dots$ — again a very close match.

Answer $\sin(0.1) \approx 0.1$, $\cos(0.1) \approx 0.995$ — both closely matching the calculator values. This illustrates the key point: the smaller θ is, the better these approximations become, because the terms being dropped (involving θ^3 , θ^4 , ...) shrink even faster than θ itself.

EXAM TIP

Applying small-angle approximations to an angle given in degrees without converting to radians first — the approximations only hold with θ measured in radians; " $\sin(0.1^\circ) \approx 0.1$ " would be numerically wrong by a factor of about 57 ($180/\pi$).

Chapter summary

Key formulas

Topic	Key formula(s)
Sine rule, cosine rule, area	$a/\sin A = b/\sin B = c/\sin C$ $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ Area = $\frac{1}{2}ab \cdot \sin C$
Radians: arc length, sector and segment area	$s = r\theta$ Sector area = $\frac{1}{2}r^2\theta$ Segment area = $\frac{1}{2}r^2(\theta - \sin\theta)$
Pythagorean identities	$\sin^2\theta + \cos^2\theta = 1$ $\tan\theta = \sin\theta/\cos\theta$ $1 + \tan^2\theta = \sec^2\theta$ $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
Addition formulae	$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
Double-angle formulae	$\sin 2A = 2\sin A \cos A$ $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
R sin/cos form	$a \sin\theta + b \cos\theta = R \sin(\theta + \alpha)$, $R = \sqrt{a^2 + b^2}$, $\alpha = \tan^{-1}(b/a)$
Small-angle approximations (θ in radians)	$\sin\theta \approx \theta$ $\cos\theta \approx 1 - \theta^2/2$ $\tan\theta \approx \theta$

Common exam mistakes across this chapter

- Calculator in the wrong angle mode (degrees vs radians) — always check before evaluating.
- Only finding the principal-value solution of a trig equation instead of every solution in the given range.
- Sign errors in $\cos(A \pm B)$, which flips compared to $\sin(A \pm B)$.
- After the R sin/cos substitution, forgetting to shift the solution range for $\phi = \theta + \alpha$ before solving, and forgetting to shift the final answers back.
- Using small-angle approximations with θ in degrees rather than radians.

You should now be able to

- Apply the sine rule, cosine rule and area formula, and work confidently in radians for arc length, sector and segment area
- Solve a trig equation over a given range, finding every solution using the graph's symmetry
- Use the Pythagorean identities and the reciprocal and inverse trig functions
- Apply the addition and double-angle formulae, and rewrite $a \sin \theta + b \cos \theta$ in R sin/cos form